

Introduction

17-313: Foundations of Software Engineering

<https://cmu-313.github.io>

Michael Hilton and Joshua Sunshine

Fall 2026

Introductions

Michael Hilton

Full Teaching Professor at CMU



A.S. Grossmont Community College 1999



B.S. San Diego State University - 2002



Software Engineer at DoD - 2002 to 2011



M.S. Cal Poly San Luis Obispo - 2013



PhD at Oregon State - 2017



Internship at Microsoft Research - Summer 2017



Assistant/Associate/Full Teaching Professor at CMU - Fall 2017 to current



Josh Sunshine

Associate Professor

B.A. Brandeis University, 2004.

Ph.D., Carnegie Mellon University, 2013

Faculty @ CMU since 2014



**Carnegie
Mellon
University**



Teaching Assistants



Autumn Qiu



Alejandro Estrada



Bing Bhakdibhumi



Cindy Wang



Anabella Chyan



Long Nguyen



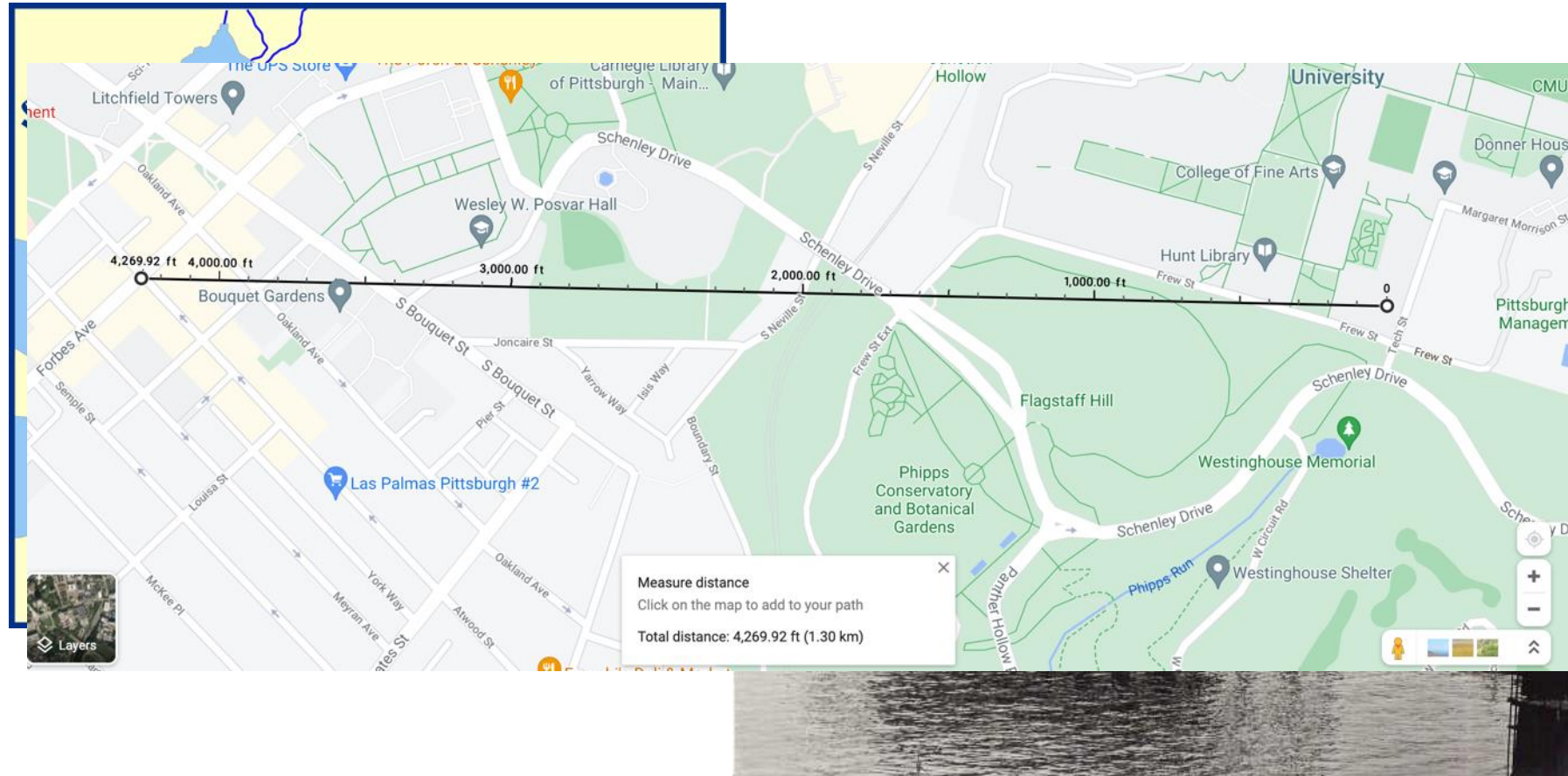
Shrey Tiwari



Vasa



Vasa



What happened is now called “Vasa syndrome”

- Changing shipbuilding orders
- No specifications for modified keel
- Shifting armaments requirements

Requirements

- Shipwright's death

Teams

- No way to calculate stability, stiffness, or sailing characteristics

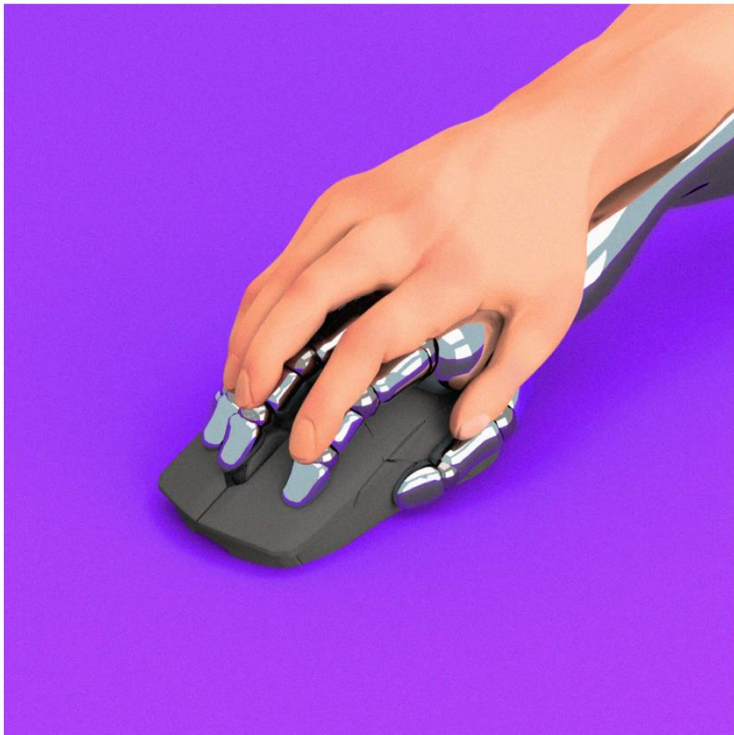
Metrics

- Failed pre-launch stability tests

QA

You're a Computer Science Major. Don't Panic.

Nov. 12, 2025



Erik Carter

By Mary Shaw and Michael Hilton

Dr. Shaw and Dr. Hilton teach software engineering at Carnegie Mellon University.

For decades, computer science students have been taught a central skill: using computers to solve problems. In practice, that has meant programming or writing code to tell a machine how to perform tasks, like sorting a list of numbers or finding the most effective way to deploy snowplows during a winter storm.

Now, generative artificial intelligence tools like ChatGPT and Claude can write those programs themselves, producing code in much the same way they write essays and legal briefs: by analyzing a vast number of similar texts and assembling new ones that look similar. A student can ask ChatGPT to write a program that sorts a list of numbers and get a working answer in seconds.

This new capability is transforming how computer scientists get work done. Microsoft's chief executive, Satya Nadella, has said that [up to 30 percent](#) of the company's code is being written by A.I. at this point. Even nonprogrammers suddenly have the ability to [create their own software tools](#).

AI is NOT killing SE Jobs

of Open Tech Jobs

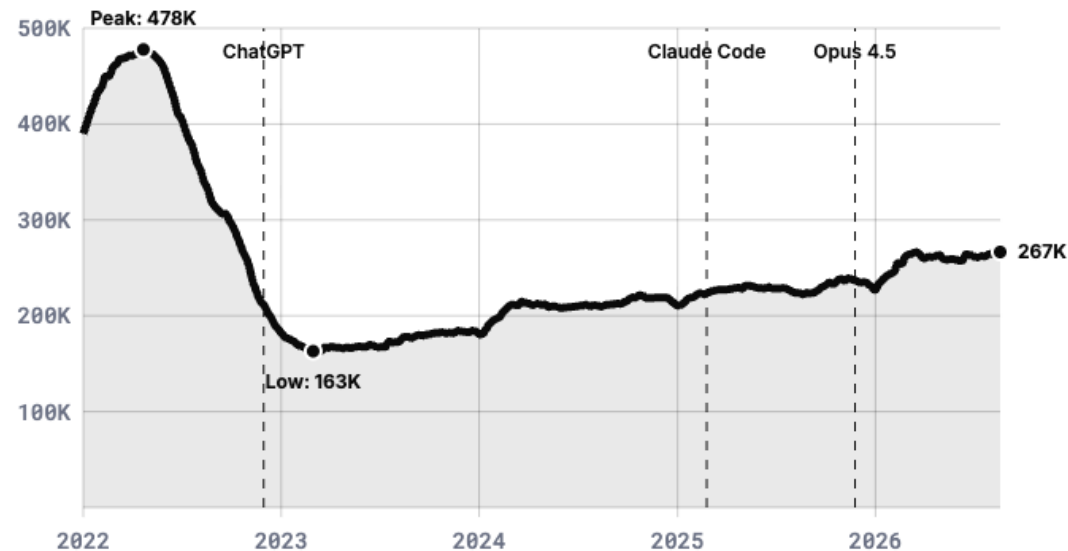
Total open jobs at public tech companies and top tech startups

266,531

As of Aug 19

↓ 44.2% from peak

↑ 63.5% from low



source: trueup.io

☐ Show U.S. Fed Funds Rates ☒ Show major tech events ☒ Label key data points

Participation Activity

- Write your name and AndrewId at the top of the paper (portrait orientation)
- Answer this question:
 - What do you hope to learn in this class?

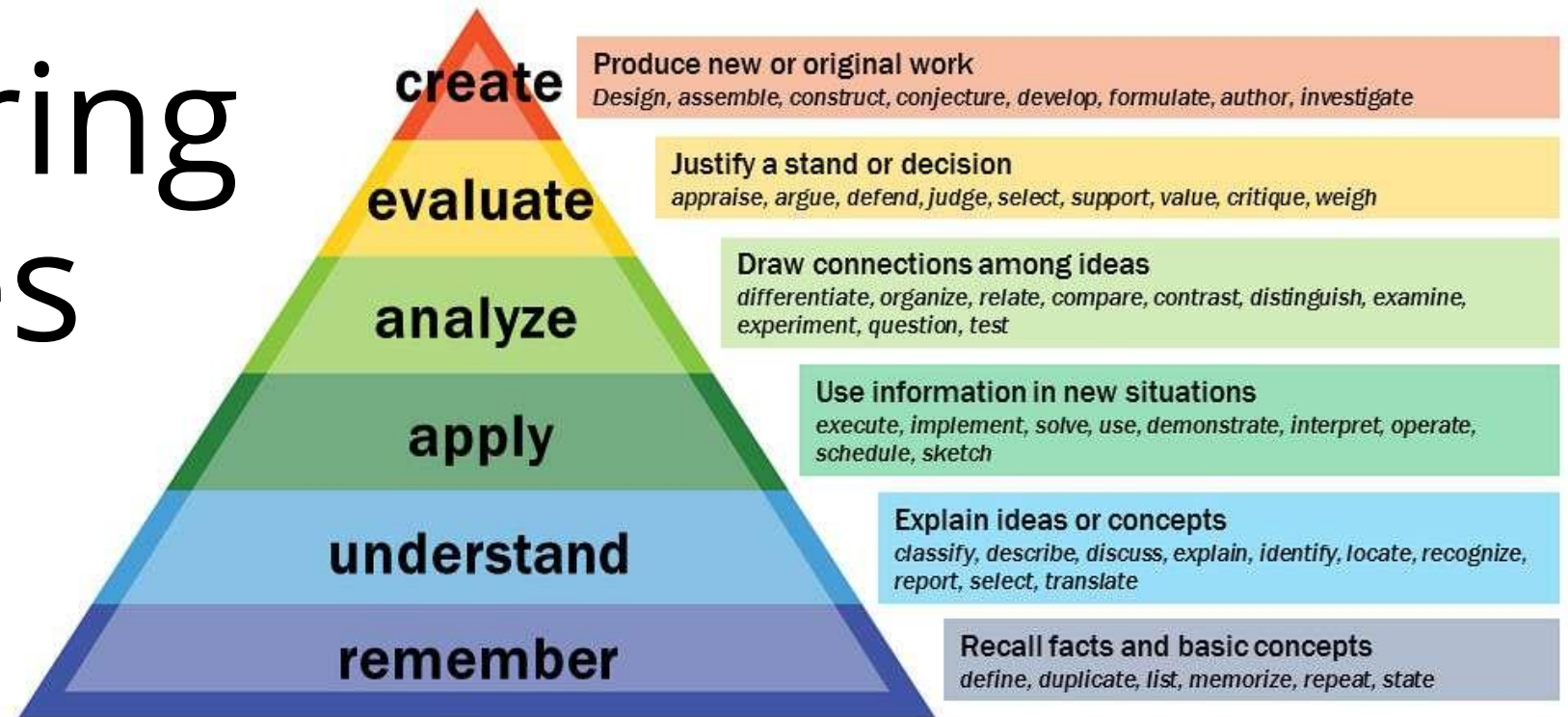
**If you
automate
your
thinking,
you
automate
away the
learning.**



Develop Engineering Judgement

Learn Engineering Principles

Bloom's Taxonomy



Vanderbilt University Center for Teaching

Course infrastructure and logistics

Recent FCE:

I might have found this course exceedingly annoying. But I can't deny the outcomes. I do feel that I've learned a lot in this class. About working in teams, with software that are hard to work with, about software development practices, technical details like Github workflows, open source projects, AI integration. When it's all said and done, I am grateful for the class and what I've taken away from it.



Laptop multitasking hinders classroom learning for both users and nearby peers

Faria Sana^a, Tina Weston^{b,c}, Nicholas J. Cepeda^{b,c,*}^aMcMaster University, Department of Psychology, Neuroscience, & Behaviour, 1280 Main Street West, Hamilton, ON L8S 4K1, Canada^bYork University, Department of Psychology, 4700 Keele Street, Toronto, ON M3J 1P3, Canada^cYork University, LaMarsh Centre for Child and Youth Research, 4700 Keele Street, Toronto, ON M3J 1P3, Canada

ARTICLE INFO

ABSTRACT

“...participants who multitasked on a laptop during a lecture scored lower on a test compared to those who did not multitask, and participants who were in direct view of a multitasking peer scored lower on a test compared to those who were not. The results demonstrate that multitasking on a laptop poses a significant distraction to both users and fellow students and can be detrimental to comprehension of lecture content.”



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No Laptops In this Class

- You may not use your laptop in this class*
- We're doing this for you AND those around you



* Unless you have an explicit accommodation that permits laptop use that is sent to us by CMU's Office of Disability Resources (ODR).

Connect with us for the class

- All links on our course website: <https://cmu-313.github.io>
- Canvas: <https://canvas.cmu.edu/courses/54889>
- Gradescope: <https://www.gradescope.com/courses/1356702>
- Slack: Check your emails (will send again if you just enroll)
 - Slack vs Piazza

Course infrastructure and logistics

- Infrastructure/source of truth
 - Course website: schedule, slides, syllabus, office hours
 - Canvas (and Gradescope) homework, grades, other material
 - Slack for communication and collaboration.
 - Git/GitHub for coding and collaboration
- Logistics
 - Lecture in-person only
 - All recitations are in-person (Except yesterday)
- Office Hours are flexible.
 - Office Hours are on website. Please come by!!
- If you want to talk to us, DM/email BOTH INSTRUCTORS at once.
 - TRUST us, it's faster. If you DM us individually, we might both assume the other instructor will reply to you first.

Course Themes

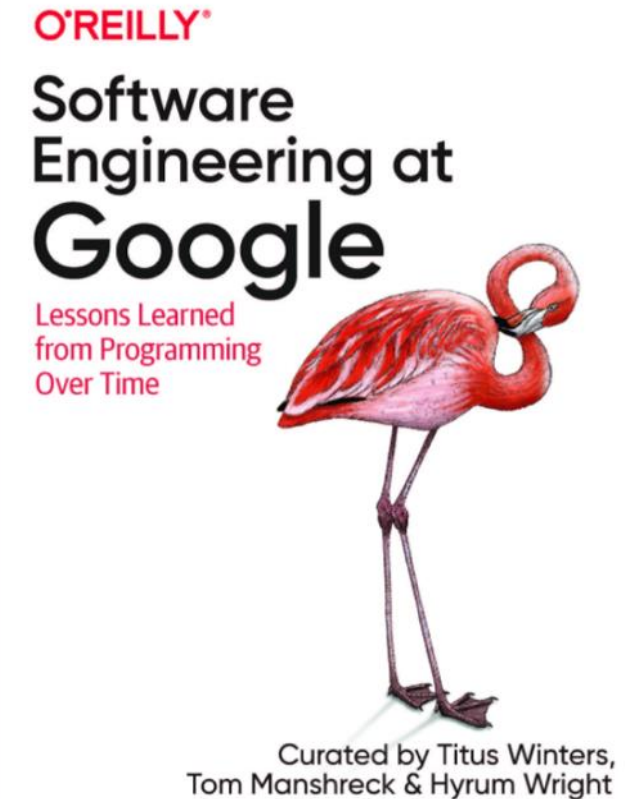
- Software engineering as a human process
- Requirements and Specifications
- Metrics and Measurement
- Software Quality: Testing + CI + Security
- Continuous Deployment and DevOps
- Software Project Teams
- Managing Time, Teams, and Risks
- Software Architecture and Design Docs
- Scaling and Performance, Trade-offs
- AI/ML in Software Engineering
- Open-Source Software
- Strategic Thinking about Software

Participation Activities

- In-person activities
 - **Lecture:** Active learning exercises in almost every lecture
 - Most will be done on physical paper and turned in at the end of class.
 - **Recitation:** Working sessions, submission on Canvas/Gradescope
- All of the above count as graded “participation activities”
 - You may miss up to 4 participation activities with no grade penalty
(No need to send emails asking permission)

Textbook

- No single textbook
- Recommended supplementary reading: Software Engineering at Google
 - Available for free online (legally!):
<https://abseil.io/resources/swe-book>



Gaining Experience: Central to 17-313!

- Case study analyses
- Team assignments
 - Teamwork exercises
- Open-source engagement
- **Hands-on experience is key!!!**
 - “Learn by Doing”

Evaluation

- Assignments (40%)
 - Regular homework projects, mostly in teams with individual component including open-source engagement (35%)
 - Problem Sets (5%)
- Midterms (30%)
- In-class Quizzes (10%)
- Participation activities (20%)
 - In-class exercises
 - Pre-class reading assignments
 - Recitation exercises

“Homework” Assignments / Projects

- P1: Individual homework installing, running, making changes to opencode.
- P2-P4: Work on opencode in a team, doing a series of SE tasks (adding features, testing, deploying, validating).
- P5: Open-source Excursion
 - Open-ended project: contribute to an OSS project using everything you have learned; get kudos for having PRs merged

Team Assignments

- Mirror realistic setting
- Assigned teams throughout the semester
 - Fill in team building survey before next lecture
- Teamwork surveys every week
- Conflict resolution process as needed
- Most team assignments have individual components

Recitations

- Practical tasks, preparation for homework, extra material, discussions
- Have your GitHub account at the ready.
 - Bring your laptop!
- This week only: Learn Typescript in an sync/offline recitation (to prepare for P1 and P2). **Don't forget to finish by Friday.**
- From P2 onward: Project teams will all be in the same recitations... Good (forced) opportunity to meet in person

Problem Sets

- Deliberate practice with key concepts of the course.
- Similar in spirit to problem sets you're probably used to in natural science or math courses.
- 5% of your grade
- Goal is to **develop** your 1) **memory** and 2) **understanding** of your **ability to apply** software engineering principles.

Quizzes

- In class, at the beginning. Don't be late!
- They will be announced in advance.
- Content will be similar or even identical to problem sets.
- 8-10 quizzes throughout the semester.
- 10% of your grade in the course.
- Goal is to **develop** your 1) **memory** and 2) **understanding** of your **ability to apply** software engineering principles.
- *If you use LLMs on the problem sets you probably won't do well on the quizzes.*

Generative AI for Coding

You may use generative AI for any coding tasks in the course. However, we believe that **your AI usage can have significant negative implications on your learning**. Our goal for you in this class is that you learn to **use AI in a way that helps you become a better software engineer**.

Generative AI for Writing

We **very strongly discourage** using AI to *Generate* writing responses, but we encourage its use for editing your writing. In addition **you are liable for factually inaccurate answers or unspecific rambling produced by AI tools**, and we will assess significant penalties to low quality (**AI Slop**) responses. It is your responsibility to validate and verify AI-produced content before submitting it for class purposes.

Professionalism

- Being a professional means, you must work well with others
- The best professionals are those who make those around them better
- If you feel someone is not treating you or someone else in a professional manner, you have two options:
 - If you feel you have the standing to do so, speak up!
 - Reach out to the course staff, and we will meet with you privately to discuss it, as well as preserve your anonymity

Final Projects

- Open-source excursion is the most fun part of the course!
- Very open-ended project. Large % of overall grade.
- Brings together everything you will have learned from lecture and prior assignments
- Teamwork and communication is very important
- In-person presentation in finals week (no exam)
- ***Do NOT book flight tickets for end-of-semester holidays until finals are scheduled.*** You will lose points for missing final presentations if you fly out early.

Late day policy

- **Assignments:** No late days
 - Simply doesn't work with team assignments
 - Plan for unexpected delays ahead of time (not just before deadline).
- **Participation activities (lecture + recitation):**
 - Everyone gets **4 free absences**. No need to inform us beforehand.
 - Beyond 4 absences, participation grade can be affected.
 - Inform us of extended absences (e.g., hospitalization). We can help you make up some of the lost points in such cases, in conjunction with your advisor.
- If you have an assignment due after a trip, turn it in *before* you leave.
 - You may not have Internet where you're going.
 - Your return travel may be delayed beyond the assignment deadline!

Academic Honesty

- In group work, be honest about contribution of group members; do not cover for others
- Let us know if you have concerns about teammate's work.
- Attribute content to any Generative AI service you use.
- DO NOT submit participation sheets for people who are not in class

For next class: survey, scheduling



First-week Survey due Thursday

- Form groups based on schedule availability
 - **This is ridiculously important.**
 - Identify experience and working styles.
 - Participation point
- Help us shape the course based on
 - your background knowledge
 - your interests
- Posting to Canvas. We will also post on Slack after inviting you all

Project P1 posted online



- **P1A due this Monday** (August 31st, 11:59pm)
 - Only 5% of total P1 points – meant to ensure you start on time
 - Only need to be able to install and run opencode locally
- **P1B due next Monday** (September 07th, 11:59pm)
 - Refactor code in a single file to address code quality warnings.
 - Validate change via tests.
 - Expect to be **technically challenging** for non-experts; purpose is to learn new things and engage with a large code base without much hand-holding.

Software engineers are now conductors of AI output. They need good engineering judgement to get good results.

